

**Cognitive and Resting-State EEG Predictors of Intensive Multilingual Learning, and
Change in Executive Function across Training: A Bayesian Multilevel Study**

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Abstract

Adults vary widely in how quickly and how well they learn a new language. We related baseline executive function (working memory, inhibitory control), statistical learning and resting-state EEG to the trajectory of grammaticality-judgement accuracy as Norwegian–English bilinguals acquired one of two artificial languages. We also tested change in executive function across training. Trial-level accuracy was modelled in a joint Bayesian multilevel framework with no per-learner derived slopes. Working memory predicted the level of grammatical knowledge learners reached, credibly so for the fully sampled gender-agreement property. Inhibitory control, statistical learning and the resting-state measures were largely uninformative, and no neural measure added predictive information beyond the behavioural ones. Working-memory span itself rose between the first and fifth sessions, so the capacity that predicted learning also changed over the training period. Attainment was estimated more reliably than learning rate, which bears on how slope-indexed aptitude findings should be read.

Keywords: second language acquisition, working memory, statistical learning, resting-state EEG, individual alpha frequency, Bayesian multilevel models, individual differences

Cognitive and Resting-State EEG Predictors of Intensive Multilingual Learning, and Change in Executive Function across Training: A Bayesian Multilevel Study

Adults differ markedly in the speed and the ultimate success with which they learn a new language, and explaining that variation is a central goal of the cognitive neuroscience of language. Three families of predictor recur in the literature. Working memory is the best-evidenced cognitive correlate of second-language outcomes, with a small-to-moderate positive meta-analytic association ([Linck et al., 2014](#)) that is stronger for executive and complex-span measures than for simple storage span. In the phonological short-term memory literature, associations with second-language vocabulary learning are likewise stronger for nonword repetition than for digit span ([Kurokawa, 2026](#)). Statistical or implicit learning, the ability to extract distributional regularities, relates positively but only weakly to language ability across 97 studies, with the size of the correlation depending on the task and the stimulus modality ([Zhou et al., 2024](#)). In one individual-differences study, nonadjacent statistical learning predicted comprehension of particular native-language sentence types ([Misyak & Christiansen, 2012](#)). Inhibitory control is the least settled. The bilingual-advantage literature finds no reliable executive-function benefit of bilingualism ([Dick et al., 2019](#); [Paap & Greenberg, 2013](#)), and direct tests of inhibition against second-language outcomes are mixed, reporting null ([Huensch, 2024](#); [Linck & Weiss, 2015](#)) as well as positive ([Chen et al., 2022](#)) associations.

Alpha and theta dynamics in the resting EEG, and the individual alpha frequency in particular, index memory and cognitive performance ([Klimesch, 1999](#)). Baseline oscillatory power has also predicted the rate of adult second-language learning directly, with faster learners distinguished mainly by higher-frequency (beta and low-gamma) rhythms ([Prat et al., 2016, 2019](#)). In the later of those studies, only the coherence measures survived correction for multiple comparisons, so the band-power precedent rests chiefly on the earlier one, together with a ten-participant pilot in which resting beta power predicted an attainment-normalised progress score in older adults ([Kliesch et al., 2021](#)). We therefore enter five canonical bands (delta, theta, alpha, beta, gamma) and the individual alpha frequency as baseline predictors. That precedent

implicates higher-frequency bands recorded over right temporoparietal cortex. Our own reduction is over posterior electrodes, so we do not expect it to transfer directly, and the bands enter without a directional prior (see *Priors*). Although intensive language learning can reshape the underlying neural architecture (P. Li et al., 2014; Mårtensson et al., 2012), the EEG here is measured only at baseline and used to predict subsequent learning.

Raw band power is, moreover, an ambiguous quantity. It conflates genuine oscillatory activity with the broadband aperiodic ($1/f$) background on which the peaks sit, and the two components carry different information (Donoghue et al., 2020). The aperiodic exponent has been proposed as a non-invasive proxy for cortical excitation–inhibition balance (Gao et al., 2017), varies systematically across individuals and with age (Voytek et al., 2015), and can drive associations that appear on the surface to belong to band power (Ouyang et al., 2020). A claim that a resting rhythm predicts learning should therefore survive the separation of the two components, and we test that directly.

A question cuts across all three families: whether an aptitude expresses itself in how fast learners improve or in the level they reach. Most individual-difference studies index learning by a per-learner slope estimated in a first stage, a design that biases the predictor effects (see *Learning outcome and model structure*). We therefore model rate and attainment jointly at trial level and ask which of the two the predictors act on.

This study brings these strands together in a controlled, longitudinal artificial-language paradigm. Norwegian–English bilinguals learn one of two miniature languages across a series of laboratory sessions. At baseline, we measure executive function (working memory via digit span, inhibitory control via the Stroop task (Stroop, 1935)), resting-state EEG and statistical learning, the last by an alternating serial reaction-time task (Howard & Howard, 1997), an alternating-sequence variant of the serial reaction-time paradigm (Nissen & Bullemer, 1987). Across the training sessions, we track the developing accuracy of grammaticality judgements. Designs in which a controlled training outcome is predicted from a baseline neural recording remain rare. Most evidence linking resting EEG to cognition is cross-sectional, and the few

longitudinal precedents (Kliesch et al., 2021; Prat et al., 2016, 2019) indexed learning by a derived rate or gain score. The present study models the full trial-level trajectory jointly, separating attainment from rate and quantifying the reliability of each. It also asks the neural question twice, of the raw band powers and of their aperiodic-adjusted decomposition, so that which characterisation of the resting spectrum the data prefer is itself treated as an empirical question.

Two questions follow. Part A asks whether the baseline cognitive and neural measures predict the rate and attainment of learning. Part B asks whether executive function changes between the first and fifth sessions, and whether any change is confined to one index or broad across the battery. Part A bears on a standing question in the individual-differences literature: whether successful adult grammar learning is underwritten by a general advantage or by a specific capacity. A general advantage should be evident across a battery of cognitive and neural measures assessed and modelled identically. A specific capacity would leave one predictor associated with attainment and its co-measured companions unrelated to it. The two accounts are separable here to the extent that every predictor enters the same joint model of the same outcome under priors of comparable scale. Identical modelling does not, however, confer comparable reliability on the measures themselves, and which comparators can bear the contrast is set out under *Limitations and future directions*. Part B has a stake of its own. Whether cognitive capacities improve with demanding training is itself contested, since apparent gains often turn out to reflect retest artefacts and expectancy (Boot et al., 2013; Simons et al., 2016). The three indices are therefore co-administered and used as internal controls on one another, because generic retest and expectancy effects act broadly across a battery.

Method

Participants

The participants are those of the companion study: Norwegian–English bilinguals (first language Norwegian, second language English) who learned one of two miniature languages. One participant was aged 16 at enrolment according to the language-history questionnaire, and all others were adults. The companion paper reports the full age distribution and the ethical approval

and consent procedures covering the sample. Of the 65 enrolled, 64 contributed grammaticality-judgement trajectories (33 Mini-English, 31 Mini-Norwegian) and 63 a baseline cognitive battery. Of these, 62 had both measures and were therefore candidates for the joint predictor model, and 57 also had usable baseline resting-state EEG. Because the joint model requires complete data on every predictor, the fitted raw-band model comprises 56 of those participants. The difference is that one participant with usable resting-state EEG had a baseline Stroop score excluded under the task's minimum-observation rule, and listwise deletion removes them. The aperiodic variant described below comprises 50 participants, the subset for which the specparam re-parameterisation is defined. That decomposition returns no individual alpha frequency for a spectrum in which it fits no alpha peak, and listwise deletion then removes those participants too. This loss is therefore not independent of the alpha measures themselves. The Session-1-versus-5 pre/post models of executive function use all participants contributing data at either session (63 at Session 1 and 54 at Session 5 for digit span, 62 and 54 for Stroop interference, and 63 and 54 for statistical learning), with the session effect estimated within participants through the by-participant intercept. Participants tested only once inform the intercept but contribute no within-participant change. Figure 1 traces attendance and the resulting analysis samples across the six sessions.

These are the samples the models are fitted to. The descriptive displays reported below are computed on wider frames, each named in the caption that carries it, and those counts are not interchangeable with these. A descriptive summary of the predictors requires only that a participant have the measure in question, not complete data on every predictor, and in places not even the usability judgement behind the 57 above. No descriptive frame in this paper is smaller than the 56 participants the raw-band model is fitted to. Of the 64 participants with a Language History Questionnaire record [LHQ3; P. Li et al. (2020)], the mean age was 25.7 years ($SD = 7.4$, range 16–54). Of that group, 40 were female, 21 male and 1 non-binary, and 2 did not disclose their gender. Most were right-handed (60 of 63 who reported handedness). English had been acquired early (age of first exposure $M = 6.8$ years) and used for many years, with high self-rated

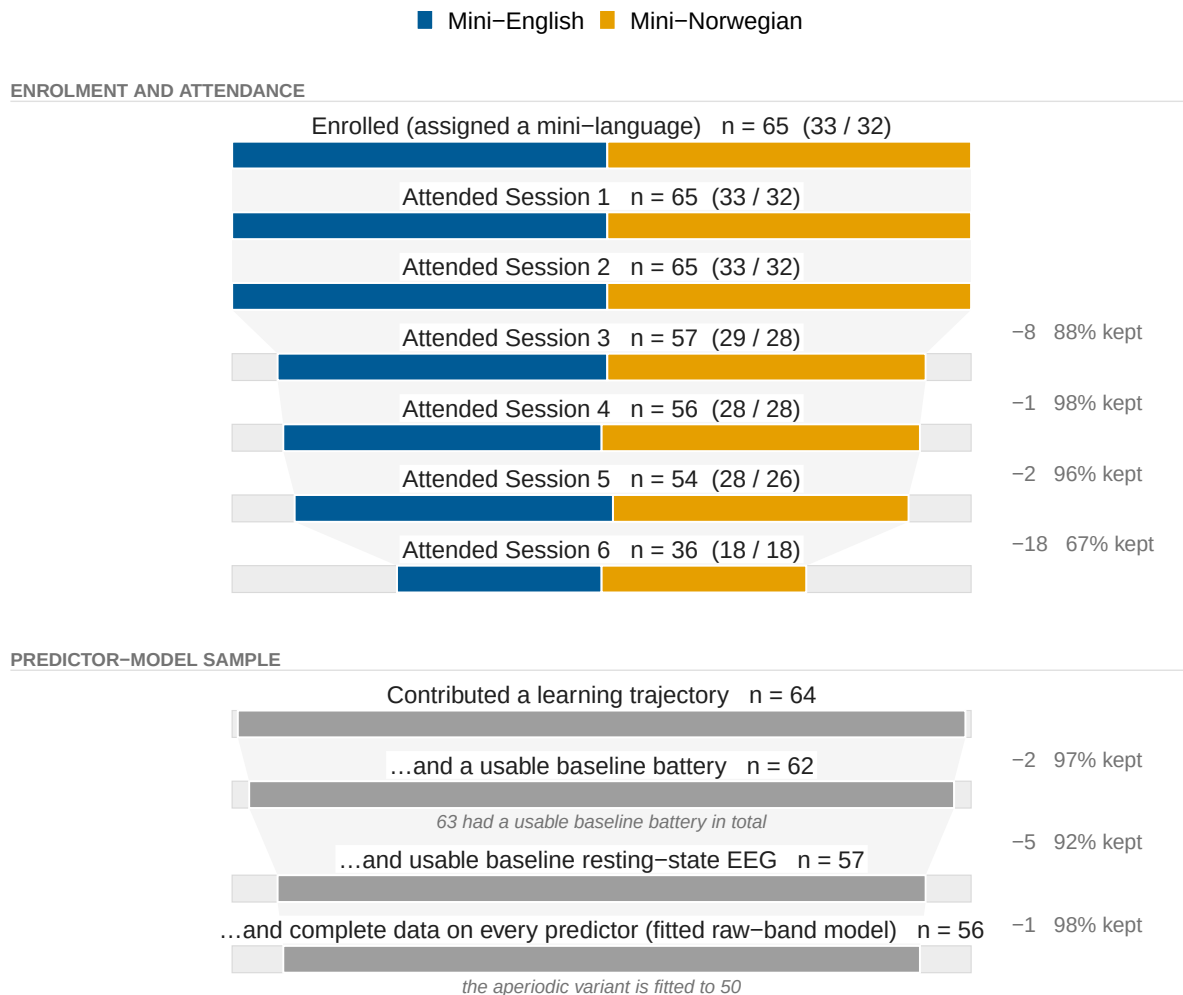
proficiency ($M = 0.82$ on a 0–1 scale). The study was conducted with institutional research-ethics approval (reference to be added) and written informed consent from all participants.

Cognitive and neural indices

The cognitive battery was administered at Sessions 1 (baseline) and 5 (post) as browser-based tasks on the Gorilla experiment platform ([Anwyl-Irvine et al., 2020](#)). Records from individuals who completed the online battery without enrolling in the study were excluded at extraction, since they have no consent linkage to the participant key. Working memory was measured by digit span (forward and backward, list lengths 3–8 with four items each, 48 scored trials, the index being the number correct). Inhibitory control was measured by a four-colour Stroop task (100 test trials, 50 of them congruent, the index being the incongruent-minus-congruent mean reaction time). Higher Stroop scores therefore index poorer inhibitory control, so a positive coefficient for this predictor would mean poorer inhibition predicting higher accuracy. Statistical learning was measured by an alternating serial reaction-time (ASRT) task ([Howard & Howard, 1997](#)), the alternating-sequence variant of the serial reaction-time paradigm ([Nissen & Bullemer, 1987](#)), of 25 blocks of 85 trials in which pattern and random trials alternate. The index is the low-minus-high-probability-triplet mean reaction-time difference, with higher scores indexing stronger learning. Reaction times were trimmed to 50–5000 ms with within-cell outliers beyond three standard deviations removed.

Resting-state EEG (rs-EEG) was recorded once, at Session 2, with a 32-channel Brain Products LiveAmp (500 Hz) in separate eyes-closed and eyes-open recordings of about five minutes each, each delimited by its own onset and offset markers. Preprocessing in BrainVision Analyzer followed the order: visual inspection of the continuous record, re-referencing to the average of the mastoids (TP9, TP10), band-pass filtering with zero-phase-shift Butterworth filters (high-pass 0.1 Hz, low-pass 30 Hz, fourth order, no notch), ocular correction by independent component analysis, and segmentation into the two resting blocks. Artefact handling was by visual inspection only, with no automatic amplitude, gradient or low-activity criteria. Essentially all marked intervals fell in the task portions of the recording and not in the resting blocks. The full

Figure 1
Participant Flow



Note. Each stage spans the full enrolled cohort. The solid block is the participants still present and the faint block completing the bar is those no longer present, so cumulative attrition reads as filled against unfilled area. Tapering bands connect consecutive stages, each transition annotated with the number lost and the percentage retained. In the upper tier, the solid block is split by mini-language. The predictor-model stages are not recorded by language and are drawn in neutral grey. The upper tier tracks attendance across all six sessions, since this paper draws on all of them: the cognitive battery at Sessions 1 and 5, the resting-state recording at Session 2, and the learning trajectory across the four judgement sessions. The lower tier is the predictor-model sample proper, which is nested, and its final stage is the sample the raw-band model is fitted to, after listwise deletion on the full predictor set. Bands are drawn only within a tier, because the analysis samples pool across sessions and are not a subset of final-session attendance. The repository's flow artefacts reproduce and cross-check every count shown.

five minutes was therefore retained for every participant in both conditions, and retention does not vary across the sample. Because no dedicated electro-oculogram electrodes were recorded, the ocular correction used surrogate channels derived from the montage. Power spectral density was estimated over eight occipito-parietal electrodes (O1, Oz, O2, P3, Pz, P4, P7, P8) by Welch's method (Welch, 1967) (2-s Hann windows, 50% overlap) and integrated into band power for delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz) and gamma (30–45 Hz). The individual alpha frequency (IAF) was the frequency of maximum power in the 7–13 Hz range (Klimesch, 1999). Corcoran et al. (2018) recommend a smoothed peak-alpha or centre-of-gravity estimator in preference to this argmax. The specparam IAF recovered in the aperiodic decomposition serves as the cross-check here.

That argmax has a failure mode that matters here, because the one suggestive neural result rests on this measure. For a spectrum that merely decays across the search range, with no peak in it, the argmax returns the lower bound of the range, so a value at 7 Hz is not necessarily a measured peak. The aperiodic decomposition resolves an alpha peak in 57 of the 60 eyes-closed recordings. For the remainder, the individual alpha frequency should be read as unresolved. No analysis excluding those recordings is reported here, so the individual-alpha-frequency association below is provisional in that respect.

The gamma band requires an explicit caveat, because it lies entirely above the offline low-pass corner. With the low-pass at 30 Hz, the whole 30–45 Hz band sits at or beyond the filter's –3 dB point. The filter's transmission was weighted by each participant's own extrapolated aperiodic background (the broadband $1/f$ component on which the oscillatory peaks sit), and the quantiles were taken across every recorded resting-state spectrum rather than across a fitted model's sample. On that basis, the retained power across that band has a median of 6.5% of what an unfiltered recording would have carried (interquartile range 6.3% to 6.6%), and no more than a median of 21.3% even under the most permissive single-pass reading of the zero-phase filter. What survives is further contaminated. No notch filter was applied, and appreciable mains energy at 50 Hz persists through the low-pass, so part of the residual band is line noise. Consistent with

that, the power observed across 30–45 Hz far exceeds what the filter would transmit of the aperiodic background (a median of 60.2% of the extrapolated background, against the 6.5% the filter would pass), so most of the residual band is not filtered neural background at all. The gamma predictor is therefore not a valid index of gamma-band activity, and we do not interpret it. We retain it in the reported models only so that the predictor set is not silently altered after the fact. Its null result should be disregarded, and on no account read as evidence that gamma-band activity is unrelated to learning.

We considered recomputing the band from the unfiltered continuous recordings, whose acquisition bandwidth of DC–131 Hz retains the frequencies of interest, and decided against it, because the resulting measure would not be interpretable either. Surface 30–45 Hz activity overlaps completely with the spectrum of cranial electromyogram, which extends from roughly 20 Hz upwards. Recordings made under complete neuromuscular blockade show differences of one to two orders of magnitude in scalp power above 20 Hz relative to the unparalysed state ([Muthukumaraswamy, 2013](#); [Whitham et al., 2007](#)). Separating neuronal from myogenic contributions in this range requires either explicit modelling of that separation or dedicated electromyogram channels ([Hipp & Siegel, 2013](#)), and this montage provides neither. Recovering the band would therefore replace a filter-attenuated measure with a muscle-dominated one, and would additionally change a pre-specified predictor after the analysis had been run. We report the band as unmeasurable in this dataset and treat a properly acquired gamma measure as a design requirement for future work, since nothing here can retrieve it.

The eyes-closed values serve as the neural predictors, since the alpha rhythm and its peak are most reliably estimated with the eyes closed. The reduction reproduces the expected posterior alpha-blocking on eye opening, which we treat as a check on the recording. Had it been absent, it would have indicated a problem with the montage or the reference before any predictor built from these data could be trusted. All continuous predictors were z -scored, so each coefficient is expressed per standard deviation of its predictor and the predictors are directly comparable with one another.

Learning outcome and model structure (Part A)

The learning outcome is the longitudinal trajectory of grammaticality-judgement accuracy, modelled jointly at trial level. The two-stage alternative is what most aptitude studies do: fit a slope per learner, then regress those slopes on the predictors. It biases the predictor effects through regression dilution, because a slope estimated from few noisy trials is treated as if it were measured without error, and it understates uncertainty for the same reason (Gelman, 2005; Humberg et al., 2024). We therefore fit a single Bayesian multilevel Bernoulli model of trial-level accuracy. The learning *rate* is a by-participant random slope for session and the predictors enter as interactions with session (Barr et al., 2013), so that under partial pooling a learner whose slope is poorly determined contributes proportionately less to the predictor estimate (Gelman & Hill, 2006):

$$\text{correct} \sim \text{Session} \times \text{Property} + \text{Session} \times (\text{WM} + \text{Inhibition} + \text{StatLearning} + \text{rs-EEG}) + (1 + \text{Session} \mid \text{participant}).$$

Session was coded as a continuous standardised variable so that the rate is a single slope. A linear effect of session is the parsimonious default. A generalised additive mixed-model (GAMM) form was held in reserve should the trajectory prove nonlinear (Sóskuthy, 2017; Winter & Wieling, 2016), since averaging across learners can distort the apparent functional form of a learning curve (Heathcote et al., 2000; Murre & Chessa, 2011). That reserve analysis was not in the end fitted, and no formal test of linearity is reported, so the session term stands as a linear approximation to the observed trajectories. Figure 2 is where its adequacy can be judged, and the gender-agreement trajectories in particular are not monotone.

We report both the learning rate (the predictor-by-session interactions) and the attainment (the model-implied end-of-training accuracy), and assess the reliability of the by-participant rate and attainment before interpreting either as an individual difference (Hedge et al., 2018; Hedge, 2021). These are single-occasion, model-based separation reliabilities estimated within this sample, computed as the variance of the shrunken by-participant posterior means relative to the

model's between-participant variance, $\text{Var}(\hat{u}_i)/\tau^2$ (by the normal-theory identity, approximately $1 - \overline{\text{Var}(u_i \mid \text{data})}/\tau^2$). They are not test-retest reliabilities, which would require a second measurement wave (Rouder & Haaf, 2019). Because session is standardised, a predictor's main effect is its association with accuracy at the average session. We label the main effects as overall accuracy and reserve *attainment* for the model-implied end-of-training quantity. Pooled accuracy is shaped by properties of differing difficulty entering cumulatively across sessions: gender agreement from Session 2, differential object marking from Session 3 and verb-object number agreement from Session 4 (see the companion paper for the design). As a sensitivity analysis, Part A is therefore additionally re-fit restricted to gender agreement, the property with full session coverage and the most stable trajectory.

Pre/post models of executive function (Part B)

Part B fits a Gaussian pre/post model of each cognitive index (Session 1 versus Session 5) with a by-participant intercept and a ± 0.5 pre/post contrast. Because the study has no no-training control group, a pre/post change is confounded with the generic threats that attend any uncontrolled retest (Shadish et al., 2002). We add a joint model across the three z -scored indices in which each acts as an internal control for the others. With digit span as the reference level, the session effect is its pre/post change and each measure-by-session term is how much less that measure changed. The model can therefore identify a change concentrated in one index and set it against the broad shift that practice, regression to the mean or expectancy would produce. The three indices are standardised within measure and enter on their recorded polarity, which is not aligned across them: higher digit span is better performance, higher Stroop interference worse. A generic improvement therefore moves the two in opposite signed directions, so the digit-span-versus-Stroop contrast confounds specificity with that opposition and should be read as an upper bound on the separation. Because resting-state EEG was recorded only at baseline, no neural pre/post is estimated. The rs-EEG measures enter Part A as baseline predictors (the rs-EEG term above).

Priors

Priors are informative and standardised on the logit scale, established from the literature reviewed in the Introduction and following a selective, evidence-tiered scheme. The cited behavioural associations are *attainment* correlations, so each small informative prior is placed on a predictor's main effect and shrunk towards zero on its interaction with session. Working memory and statistical learning receive small positive priors (Kurokawa, 2026; Linck et al., 2014; Misyak & Christiansen, 2012; Zhou et al., 2024). Inhibitory control receives a near-zero regularising prior, reflecting its weak-to-null evidence base (Chen et al., 2022; Huensch, 2024; Linck & Weiss, 2015), and the resting-state EEG predictors receive near-zero regularising priors. Although the precedent points to a higher-frequency advantage (Prat et al., 2016, 2019), those effect sizes were obtained over a different scalp region and for a different outcome, learning rate. The bands are therefore treated as an extension of that question to a new region and outcome, and no directional prior is encoded. The consequence is asymmetric: a credible band-power effect over posterior sites would have counted for the precedent, but a null obtained under regularising priors at this sample size cannot count against it. Variance components receive generous half-Student- $t(3, 0, 2.5)$ priors and correlations an LKJ(2) prior (Gelman, 2006; Lewandowski et al., 2009). In a sensitivity analysis, the Part A predictors were re-fit under weakly informative priors. As intended, the regularising priors on the resting-state bands shrank them towards zero and tightened their intervals (across all coefficients, the weakly informative intervals were about 54% wider at the median). No band-power predictor was credible under either prior. The individual alpha frequency, however, is prior-sensitive. Under the weakly informative prior, its gender-agreement interval excluded zero ($b = 0.47$, 95% CrI [0.01, 0.94]), whereas under the regularising prior it did not ($b = 0.27$, 95% CrI [-0.04, 0.58]). The regularised estimate is treated as primary, and we read the association as suggestive. The working-memory main effect was, if anything, larger under the weakly informative prior, and its pooled interval then excluded zero ($b = 0.418$, 95% CrI [0.003, 0.836]). The predictor-by-session interactions likewise remained non-credible under weakly informative priors (largest $p_d = .96$), so the attainment-not-rate

asymmetry reported below is a property of the data and does not depend on the prior specification. The Part B pre/post models use weakly informative priors throughout (Depaoli & van de Schoot, 2017; Schad et al., 2021). The weakly informative sensitivity fits place $\mathcal{N}(0, 1.5)$ on the intercept and on every population-level coefficient, with the variance and correlation priors unchanged.

The full specification is as follows, so that the models can be rebuilt from the manuscript alone. In the Part A models, the intercept has a $\mathcal{N}(1, 1)$ prior on the logit scale (judgement accuracy expected above chance) and any population-level coefficient without a term-specific prior has $\mathcal{N}(0, 1)$. By-participant standard deviations have half-Student- $t(3, 0, 2.5)$ and the random-effect correlation matrix LKJ(2). The term-specific priors are, for the main effect and its interaction with session respectively: working memory $\mathcal{N}(0.10, 0.25)$ and $\mathcal{N}(0.05, 0.15)$; statistical learning $\mathcal{N}(0.10, 0.20)$ and $\mathcal{N}(0.05, 0.15)$; inhibitory control $\mathcal{N}(0, 0.15)$ and $\mathcal{N}(0, 0.10)$; and every resting-state predictor, whether a raw band power, the individual alpha frequency or a quantity derived from the specparam spectral decomposition described below, $\mathcal{N}(0, 0.25)$ and $\mathcal{N}(0, 0.15)$. The Part B models use $\mathcal{N}(0, 1)$ on the intercept and on coefficients, and half-Student- $t(3, 0, 2.5)$ on both the by-participant standard deviation and the residual standard deviation. The joint model, whose by-participant structure includes a random pre/post slope, additionally places an LKJ(2) prior on the random-effect correlation.

Sampler, diagnostics and inference

Models were fitted with brms (Bürkner, 2017, 2018) as a front end to Stan (Carpenter et al., 2017), using the same sampler configuration as the companion paper: four chains of 10,000 iterations each, the first 2,000 discarded as warm-up, giving 32,000 post-warm-up draws, with `adapt_delta = 0.95` and `max_treedepth = 12`. Sampling used within-chain `reduce_sum` threading at two threads per chain. Because `reduce_sum` takes the grouping of its partial sums from the thread count, the recorded draws depend on that setting as well as on the seed. The convergence gate was rank-normalised $\hat{R} < 1.01$, bulk- and tail-ESS above 400, and zero divergent transitions (Vehtari et al., 2021). Posterior predictive checks were written to the repository at fit time (Gabry et al., 2019), and inference was by posterior medians, 95% credible intervals and the

probability of direction (p_d) (Kruschke & Liddell, 2018; Makowski et al., 2019). All credible intervals are equal-tailed quantile intervals, bounded by the 2.5th and 97.5th percentiles of the posterior draws. An ESS of 400 is the threshold at which the $\hat{R}$ and ESS diagnostics are themselves trustworthy (Vehtari et al., 2021). It is a weaker guarantee than the stability of an interval *limit*, which is estimated far less efficiently than a posterior mean, so borderline coefficients should be read with that in mind (Kruschke, 2021). As in the companion paper, we describe an effect as credible when its 95% credible interval excludes zero, and otherwise read the probability of direction as a continuous index of directional certainty. We do not treat it as a dichotomous test. The projection-predictive and aperiodic analyses narrow the predictor space. We apply no blanket multiplicity correction, which would inflate Type II error without an inferential gain (Rubin, 2021). Binary accuracy is modelled with a Bernoulli (logit) likelihood throughout. Analyses were carried out in R [*version pending a pipeline run*] (R Core Team, 2025) with brms [*version pending a pipeline run*] as a front end to CmdStan [*version pending a pipeline run*], projpred [*version pending a pipeline run*] for the projection-predictive selection and loo [*version pending a pipeline run*] for the Pareto-smoothed importance-sampling leave-one-out (PSIS-LOO) comparison, under the fixed seed 20260607, in a reproducible here()-anchored compendium (Marwick et al., 2018). The models were fitted on a Linux high-performance computing cluster ([*platform pending a pipeline run*]), where the software environment is supplied by environment modules and not by a container. The versions given above are therefore those of the fitting run, and not necessarily those of the machine that renders this document. The Stan models were compiled with GCC 14.2.0, the compiler the R module supplies, at an optimisation level pinned to -O1 in CmdStan's make/local, because that compiler version raises an internal compiler error on Stan's reduce_sum templates at -O2 and above. The pin governs whether the models compile and not the draws they produce. The repository's cluster environment script applies it automatically, so a rebuild on the same cluster inherits it and a rebuild elsewhere under the same compiler needs it set by hand. Figures are drawn with ggplot2 (Wickham, 2016) using the theme and colour system of depictr (Bernabeu, 2026), whose default qualitative

palette is the colourblind-safe Okabe–Ito set. The companion paper uses the same system, so the two share one visual identity.

Predictor selection and a $1/f$ -adjusted robustness check

Two further analyses probe the neural predictors. First, raw resting-state band power conflates genuine oscillatory activity with the aperiodic $1/f$ background, and that background is itself a substantive individual difference (Donoghue et al., 2020; Ouyang et al., 2020). We re-parameterised each participant’s baseline resting spectrum into an aperiodic component (exponent and offset), $1/f$ -adjusted oscillatory (theta, alpha, beta) power and the specparam-estimated individual alpha frequency, using a specparam decomposition (Donoghue et al., 2020), and re-fit Part A with these de-confounded quantities in place of the raw bands. The decomposition is our own R implementation of that algorithm and does not call the Python `foooof` package. It was fitted over 2–40 Hz with at most six peaks, peak widths constrained to 1–12 Hz, a minimum peak height of $0.05 \log_{10}$ power above the flattened spectrum, a peak threshold of two standard deviations of that spectrum, and the aperiodic component in fixed (knee-free) mode. This aperiodic variant is a robustness check of the raw-band model, motivated by the periodic/aperiodic conflation. It is not a response to the raw-band result, and it does not amount to a second independent pass over the predictor space. Like the rest of the analysis plan, it was specified after data collection (see *Transparency and preregistration*). The two reference models are compared by leave-one-out cross-validation (Vehtari et al., 2017), so the choice between the two neural characterisations rests on a quantified comparison.

A trial-level criterion cannot settle that comparison on its own. The two reference models differ only in participant-level resting-state predictors, which are constant within a participant. With about 529 trials per participant in the pooled stratum, a held-out trial’s by-participant intercept is already pinned by that participant’s remaining trials, so a between-participant predictor can add little to predicting it. Leave-one-*trial*-out PSIS-LOO is therefore precise, but about a quantity that is not the one at issue, and it would return a near-zero difference for informative and uninformative resting-state predictors alike. Blocking the folds on the grouping

factor is the recommended remedy wherever such dependence exists, even when the fitted model already accounts for the dependence with a random effect, as ours does (Roberts et al., 2017). We therefore also compare the models by K-fold cross-validation with folds blocked on participants ($K = 10$, the same fold assignment for both models, participants dealt round-robin over a seeded shuffle). We report the difference in expected log predictive density (elpd) with a standard error clustered on participants. The blocking follows Roberts et al. (2017) and the predictive score Vehtari et al. (2017). The clustered standard error is our own construction. Because the pointwise contributions are not independent within a participant, we sum them within participant before taking the standard error across participants. A difference is read as decisive only when it exceeds twice this clustered standard error, and the naive independent-observation standard error is reported alongside so the size of the clustering correction is visible.

Second, the joint Part A model deliberately retains every candidate predictor for inference, but one also wants to know which subset accounts for its predictive performance. We answer this with projection-predictive variable selection (Piironen et al., 2020; Piironen & Vehtari, 2017), which projects the rich reference posterior onto nested submodels and ranks predictors by the predictive performance retained after projection. Each submodel inherits the reference model’s regularisation and is not re-estimated from scratch. Because the reference model is a multilevel Bernoulli model, we use the latent projection (Catalina et al., 2021), the appropriate and recommended projection for this model class. The forward search is cross-validated in full, with every selection step re-run within each fold, to pay for the selection-induced optimism of naive fit-then-select pipelines. The folds are formed by participant, so that no participant’s correlated trials straddle the train/test split (Roberts et al., 2017). Ten such participant-grouped folds were used, with twenty clustered posterior draws during the search and 400 draws for performance evaluation, which was carried out on the response (0/1) scale under the latent projection. The suggested submodel size is projpred’s standard heuristic: the smallest size whose cross-validated elpd lies within one standard error of the reference model’s (Piironen et al., 2020). A predictor that the minimal submodel does not retain is not thereby shown to be null. The selection

summarises which predictors account for the reference model's predictive performance, and the full Part A posterior remains the inferential statement (Piironen et al., 2020). The raw-band and aperiodic reference models' selections are reported side by side.

Transparency and preregistration

The study design, materials and session protocol were preregistered before data collection (<https://osf.io/tjr54>). The preregistration fixed the design, the tasks and the six-session schedule but did not specify a statistical analysis plan. The Bayesian multilevel models, the informative priors, the specparam/aperiodic re-parameterisation and the projection-predictive selection reported here were therefore specified after the data were collected, and they are exploratory rather than confirmatory (Nosek et al., 2018; Wagenmakers et al., 2012). The preregistration also motivated the Session-1 battery as an index of attention relevant to the selection of a source of transfer. The present study reframes the same three tasks as working-memory, inhibitory-control and statistical-learning measures, used both as predictors of learning and as pre/post outcomes. That re-conceptualisation is a departure from the registered rationale and is likewise exploratory.

Data collection was preregistered for two sites, Norway and Spain. The present analyses concern only the Norwegian cohort (first-language Norwegian, second-language English). The Spanish-site data are not analysed here, and the findings generalise only to first-language-Norwegian learners. Finally, the preregistered protocol required participants to score above 80% on a post-training behavioural test in each grammar session (with a second attempt allowed) before they could proceed. This performance-conditional gate shapes which participant-sessions contribute learning data and is part of the attrition described above.

Results

Convergence diagnostics for every model reported below are given in Table A1, in the appendix, as realised values, since a realised value distinguishes a model that converged comfortably from one that converged marginally. All 11 models were assessed against the same gate and 11 passed it, with rank-normalised $\hat{R}$ below 1.01, effective sample sizes above 400 in both the bulk and the tail, and no divergent transitions.

Descriptive overview

Before the models, we show the observed data they summarise. Figure 2 plots the raw learning trajectory, which is also the direct check on the linear-in-session assumption the models make. Figure 3 shows the baseline predictors' intercorrelations and Table 1 gives the same predictors numerically. The per-measure distributions (Figure C1) and the resting-state spectra behind the band-power and individual-alpha-frequency measures (Figure C2) are given in the online supplementary material (Appendix C), which is deposited with the analysis code in the project's OSF archive. These displays cover three different frames, as their notes state. Figure C1 plots the stored records for each measure one panel at a time, so the number of points differs between panels. Figure 3 and Table 1 cover the participants who have both a baseline cognitive battery and an eyes-closed resting-state spectrum, and Figure C2 covers every recorded spectrum. All three are wider than the 56 participants of the fitted raw-band model, so their counts describe the measures rather than the models.

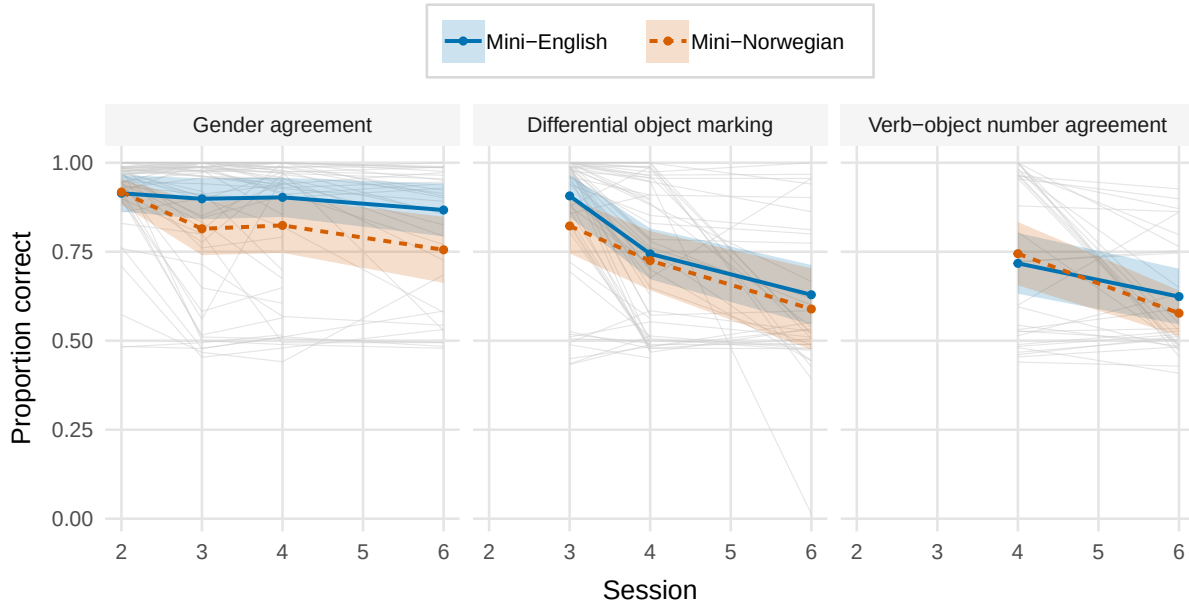
One feature of the learning trajectory bears on how *rate* and *attainment* should be read, since both terms would otherwise suggest improvement. Mean accuracy is highest at each property's first session and lower at its last, in every property and both mini-languages. The session term is credibly negative both pooled ($b = -1.16$, 95% CrI $[-1.37, -0.96]$, $p_d = 100.0\%$) and within gender agreement ($b = -0.41$, 95% CrI $[-0.56, -0.26]$, $p_d = 100.0\%$). In the pooled model, this partly reflects the design: properties of increasing difficulty enter cumulatively across sessions, so later sessions sample harder material (see *Learning outcome and model structure*). Within gender agreement, which is sampled throughout, that explanation does not apply, and we record the decline as an unexplained feature of these data. Throughout what follows, *rate* is the slope of this trajectory and *attainment* its model-implied end point, whichever way the trajectory runs.

Part A: predictors of the learning trajectory

Each predictor enters twice: as a main effect, indexing the overall level of accuracy, and in interaction with session, indexing the learning *rate*. Because session is standardised, the main

Figure 2

Observed Grammaticality-Judgement Accuracy Across Sessions, by Grammatical Property and Mini-Language

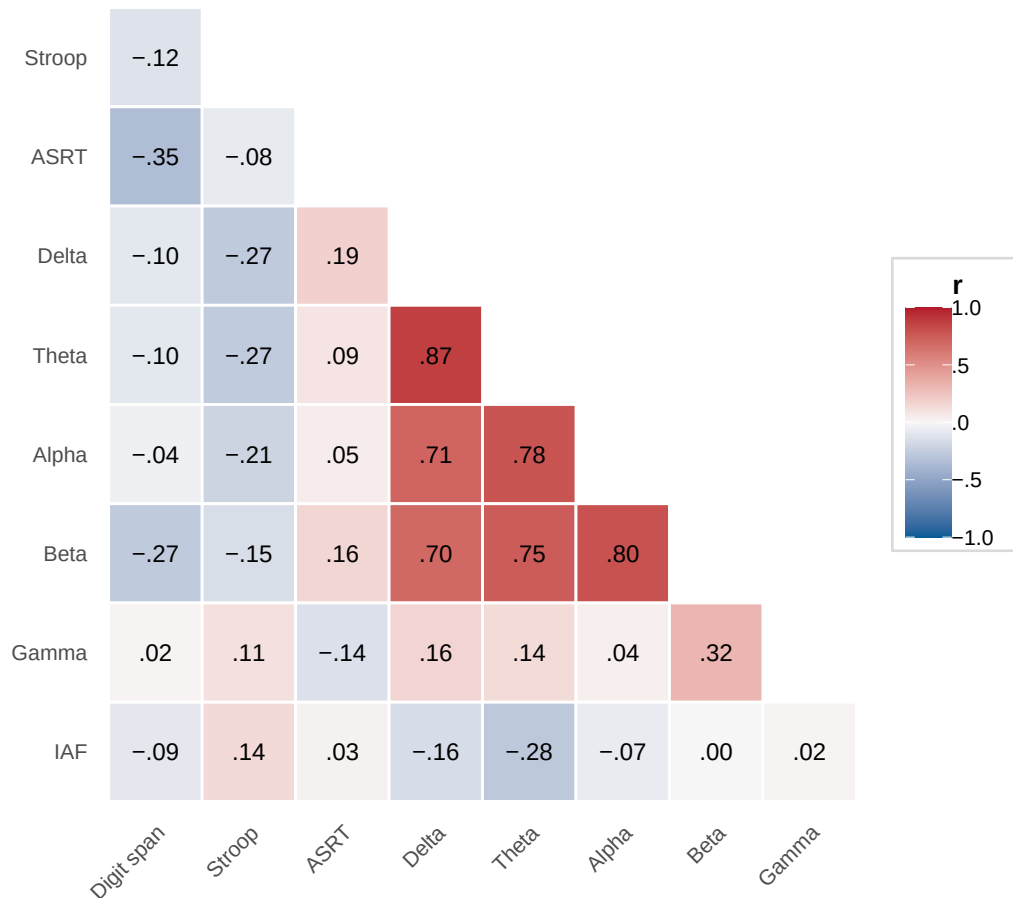


Note. Thin lines are individual participants, and the ribbon is a 95% confidence interval computed between participants, the sampling unit being participants and not trials. This is the raw trajectory the joint model summarises, and the direct check on the linear-in-session assumption.

effect is the association at the average session (see Method). Working memory was the clearest predictor. Higher baseline digit span went with higher overall accuracy, credibly so for the fully sampled gender-agreement property ($b = 0.43$, 95% CrI [0.09, 0.74], $p_d = 99.5\%$) and positively, though with an interval that narrowly included zero, in the pooled model ($b = 0.290$, 95% CrI [-0.004, 0.563], $p_d = 97.3\%$). Inhibitory control ($b = 0.06$, 95% CrI [-0.17, 0.28], $p_d = 68.9\%$) and statistical learning ($b = 0.01$, 95% CrI [-0.25, 0.27], $p_d = 53.4\%$) were not credibly related to accuracy, in line with their weak-to-null evidence base.

Of the baseline resting-state measures, the band powers were uninformative (Figure 4), but that null must be read against how strongly they are intercorrelated. Entering the bands simultaneously means each coefficient is the unique contribution of that band with the others held constant, and the four low-frequency bands share a great deal of variance (Figure 3). Their variance inflation factors are 4.5 to 6.8, which widens their posterior intervals by roughly 2.1 to

Figure 3
Pearson Correlations Among the Baseline Predictors



Note. Correlations are computed on the z-scored predictors, over the participants who have both a baseline cognitive battery and an eyes-closed resting-state spectrum, pairwise-complete, so the n behind each cell varies slightly. Only the lower triangle is shown, the redundant mirror and the unit diagonal being omitted. The resting-state band powers are expected to be positively intercorrelated through the shared aperiodic $1/f$ background, which motivates the specparam robustness check and is why the projpred ranking must be read against this structure.

Table 1*Descriptive Statistics for the Baseline Predictors*

Predictor	n	M	SD	Mdn	Min	Max
Working memory: digit span (items correct)	58	25.86	7.03	25.00	10.00	43.00
Inhibitory control: Stroop interference (ms)	57	116.27	76.51	101.63	-41.72	306.64
Statistical learning: ASRT difference (ms)	58	44.65	55.14	15.04	-2.38	198.43
Delta power (μV^2)	58	8.88	5.92	6.91	2.34	31.07
Theta power (μV^2)	58	9.03	11.42	4.62	1.33	65.16
Alpha power (μV^2)	58	48.28	49.85	30.66	1.29	224.41
Beta power (μV^2)	58	9.52	6.14	8.09	2.26	37.60
Gamma power (μV^2 , not interpreted)	58	0.66	0.52	0.50	0.14	3.10
Individual alpha frequency (Hz)	58	10.18	1.09	10.50	7.00	12.00

Note. The frame is the participants who have both a baseline cognitive battery and an eyes-closed resting-state spectrum, which is wider than the sample any Part A model is fitted to, and the n column gives the number contributing to each row. Each row carries its own unit. Because a band power is the spectral density integrated over the band, its unit is squared microvolts and not squared microvolts per hertz.

2.6 times relative to an orthogonal predictor. Their null coefficients are therefore weakly identified and uninformative about whether those rhythms relate to learning, and they are not directly comparable to the zero-order band–outcome associations reported in the precedent literature. Two of the neural predictors escape this: gamma (VIF 1.51) and the individual alpha frequency (VIF 1.29) are close to orthogonal to the rest, so their estimates stand on their own. The one suggestive signal was the individual alpha frequency, whose association with gender-agreement accuracy was $b = 0.27$, 95% CrI $[-0.04, 0.58]$, $p_d = 95.3\%$. The cross-check the Method promised points the same way without settling it. In the aperiodic re-parameterisation of the same stratum, fitted to a smaller sample, the specparam-estimated individual alpha frequency was $b = 0.24$, 95% CrI $[-0.10, 0.57]$, $p_d = 91.7\%$, and the aperiodic exponent, the quantity the decomposition adds, was $b = 0.07$, 95% CrI $[-0.30, 0.43]$, $p_d = 64.7\%$. The learning-rate interactions were uniformly weak (for working memory, $b = -0.10$, 95% CrI $[-0.27, 0.07]$, $p_d = 87.6\%$), consistent with priors that placed their small informative mass on attainment and shrank the rate terms towards zero.

To establish which predictors account for the reference model’s predictive performance,

projection-predictive selection was run on both the pooled and the gender-agreement reference models and on the aperiodic re-parameterisation, and it returned the same parsimonious answer throughout. In the raw-band model, the suggested size was 6: the design scaffold (by-participant intercept and session slope, session, grammatical property and their interaction) together with working memory (digit span). Digit span was the first individual-difference predictor selected, entering at step 6 of the forward search, exactly the suggested size. Inhibitory control (step 8), statistical learning (step 10) and the resting-state band predictors entered only beyond the suggested size, so they are not among the terms the search retained. The forward search was capped at projpred's default path length, so the lowest-ranked terms, including the individual alpha frequency, reported above as a suggestive baseline signal, fell below that cap and were not positively ruled out in that stratum.

The gender-agreement stratum, fitted to that property alone, gives the same answer: a suggested size of 4, with working memory again the first individual-difference predictor to enter, at step 4, and inhibitory control and statistical learning entering beyond it (steps 6 and 8). That search ran over the full candidate set rather than projpred's default path length, so gamma power and the individual alpha frequency were evaluated and not truncated away, and they rank last of all (steps 18 and 20). The full-length search was run only for this stratum, the one the individual-difference claim rests on. There, neither term added anything the retained predictors had missed, and how far beyond the suggested size a term lands is not interpreted.

The aperiodic re-parameterisation reproduced this ordering (suggested size 6), with the aperiodic exponent and offset entering at steps 12 and 14, well beyond the suggested size, so the $1/f$ background added no predictive information the raw bands had missed. That search was capped at the same default path length, so the adjusted beta power and the specparam individual alpha frequency were not reached in it either, and the same caveat applies to them as to the raw model's tail terms. This is a statement about predictive parsimony, not evidence that the unselected predictors have no effect ([Piironen et al., 2020](#)).

The two parameterisations were also compared by PSIS-LOO on the sample they share.

Because the specparam re-parameterisation is defined for 50 of the 56 participants, the raw-band model was refit on that common subsample so the two are compared on identical observations. The aperiodic-minus-raw difference in expected log predictive density was 0.2 (SE 0.5) in the pooled stratum and -0.1 (SE 1.0) for gender agreement, in each case smaller than twice its standard error (Vehtari et al., 2017). As noted in the Method, this leave-one-trial-out criterion cannot detect participant-level predictors, so the near-zero difference is not read as evidence that the two characterisations are equivalent.

We then ran the participant-grouped comparison directly, refitting both models on the common sample under the 10-fold participant-blocked scheme described in the Method. The aperiodic-minus-raw difference was -56.5 in the pooled stratum and -20.9 for gender agreement. These are reported with standard errors clustered on participants (51.3 and 31.4): the usual standard error treats the pointwise contributions as independent, which under participant-blocked folds they are not, and doing so understates the uncertainty here by factors of 6.5 and 4.3. On that corrected scale, neither difference reaches twice its standard error, so neither characterisation predicts a held-out participant's trajectory detectably better than the other. Both differences favour the raw bands in direction, but neither is distinguishable from zero, and we draw no conclusion from the sign. The comparison is also blunt: at these clustered standard errors only a difference beyond about 103 nats in the pooled stratum and 63 for gender agreement would be read as decisive, so this is a tie that bounds the comparison rather than one that establishes equivalence. That is the same verdict the trial-level comparison returned, now reached on the criterion the question calls for.

Model-implied end-of-training attainment was 0.63 (95% CrI [0.57, 0.70]) pooled and 0.88 (95% CrI [0.83, 0.92]) for gender agreement. The by-participant level at the average session (the intercept) was estimated with high reliability (pooled .84, gender-only .79). The learning-rate slope was estimated less reliably (pooled .75, gender-only .42), and because the gender-only rate is the least reliably estimated of these quantities we do not interpret rate-based individual-difference claims from that model.

Table 2 gives these outcome reliabilities in full, and Table B1 the internal consistency of the three baseline cognitive indices. Both matter, because the association between a predictor and an outcome is bounded by the reliability of each. Two of the three cognitive indices are moreover difference scores, a form whose reliability is typically lower than that of its components (Draheim et al., 2019; Hedge et al., 2018).

Table 2

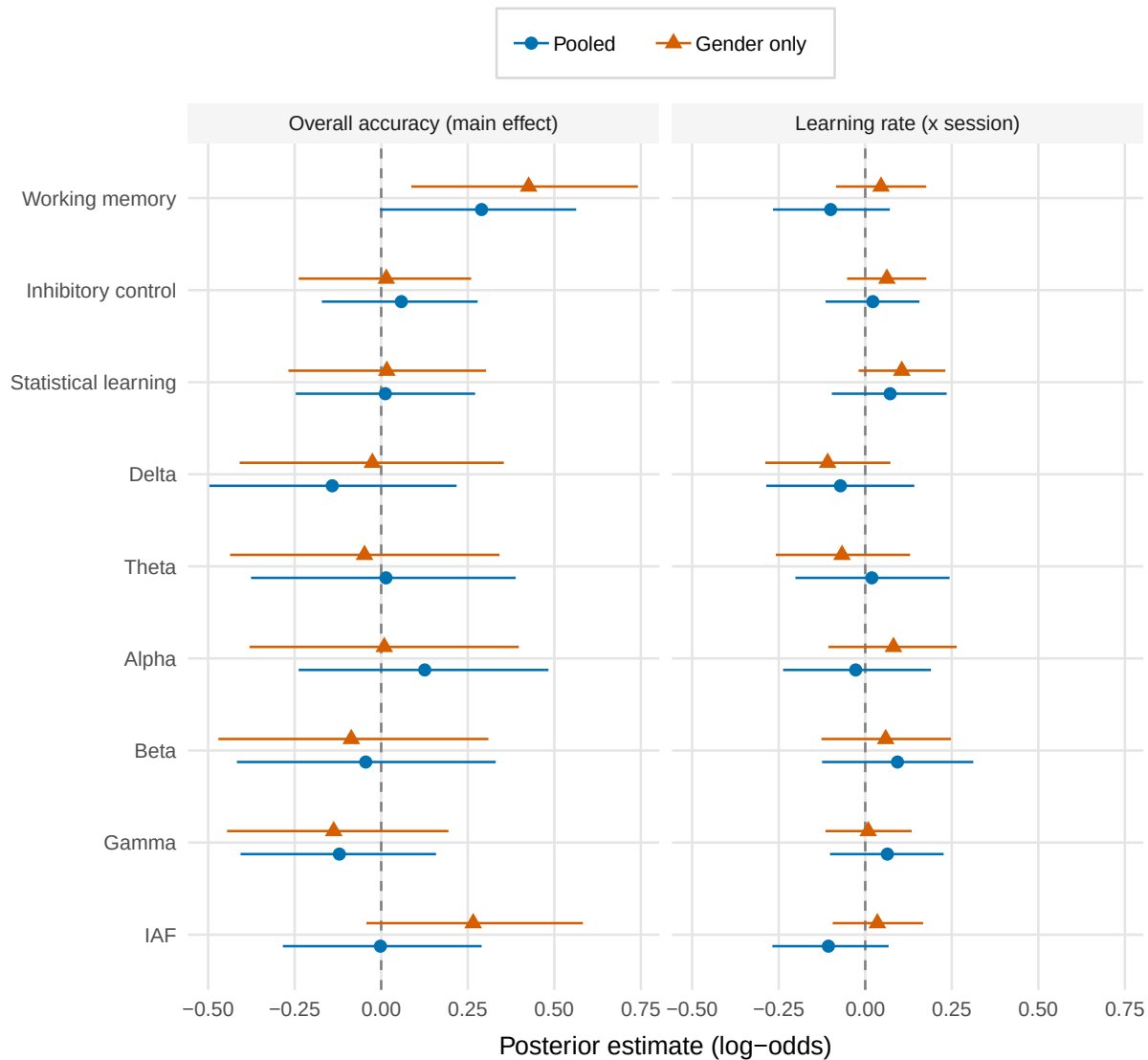
By-Participant Reliability of the Learning Trajectory

Model	Quantity	Group SD	Mean posterior SE	Reliability
Pooled	Overall accuracy (intercept)	1.30	0.47	.84
Pooled	Learning rate (session)	0.71	0.33	.75
Gender only	Overall accuracy (intercept)	1.47	0.59	.79
Gender only	Learning rate (session)	0.39	0.28	.42

Note. Columns give the group standard deviation, mean posterior standard error, and reliability of the intercept (a participant's level at the average session, since session is standardised) and of the session (rate) slope, for the pooled and gender-agreement-only models. The reliability column is the empirical-Bayes form, the variance of the shrunken by-participant posterior means over the model's between-participant variance. The group SD and mean posterior SE columns are shown for context, not as the terms of a formula.

Part B: change in executive function over training

Comparing the first and fifth sessions, working-memory span increased over training, $b = 0.29$, 95% CrI [0.09, 0.48], $p_d = 99.8\%$. The Stroop interference effect ($b = -0.14$, 95% CrI [-0.49, 0.22], $p_d = 78.4\%$) and the statistical-learning effect ($b = -0.13$, 95% CrI [-0.47, 0.21], $p_d = 78.4\%$) showed no credible change. In the joint model that lets each index control the others, the pattern pointed the same way. The joint-model digit-span change was $b = 0.29$, 95% CrI [-0.05, 0.64], $p_d = 95.3\%$, an interval that, unlike the single-index model's, includes zero. The estimate is less precise in the joint specification, which shares one residual standard deviation across the three measures and adds a by-participant change slope. Relative to that change, the pre/post shift was smaller for the Stroop interference effect ($b = -0.43$, 95% CrI [-0.92, 0.06], $p_d = 95.7\%$) and for the statistical-learning effect ($b = -0.42$, 95% CrI [-0.92, 0.05], $p_d = 95.7\%$). These specificity contrasts have credible intervals that include zero (probabilities of

Figure 4*Part A Predictor Effects on Overall Accuracy and Learning Rate*

Note. Points are posterior medians with 95% equal-tailed credible intervals for each baseline predictor's association with overall accuracy (the main effect, the association at the average session) and with learning rate (the predictor-by-session interaction), in the pooled and gender-agreement-only models. Estimates are on the logit scale: main effects are the change in log-odds of a correct response per SD of the predictor, interaction terms the change in the per-session slope per SD. The two panels share a common x axis so the families are directly comparable, and the rate-interaction terms carry deliberately tighter priors than the main effects. Predictors are ordered working memory, inhibitory control and statistical learning (for the Stroop-based inhibitory-control predictor, higher scores mean poorer inhibition), then the resting-state measures. Intervals excluding zero indicate effects whose direction is well determined.

direction 95.7% and 95.7%), so the evidence that the gain is working-memory-specific is suggestive without being decisive. Figure 5 shows these effects.

Two features of the comparison qualify it further. First, the three indices are z -scored within measure and enter the joint model on their recorded polarity, and that polarity is not aligned across them. A higher digit span is better performance, a higher Stroop interference score worse (see *Cognitive and neural indices*). A generic practice improvement would therefore push digit span up and Stroop interference down, in opposite signed directions, so the digit-span-versus-Stroop contrast is in part an artefact of that opposition, and it overstates the separation correspondingly. The ASRT comparison is not affected in the same way, since a higher ASRT score also means better performance and its polarity therefore matches digit span's. Second, the Stroop index has very low internal consistency here (see *Limitations and future directions*), which leaves it a noisy control. In task units, the digit-span change was from 26.0 items correct at Session 1 to 28.4 at Session 5, against 112.6 to 99.3 ms of Stroop interference and 43.0 to 35.5 ms of ASRT learning.

Discussion

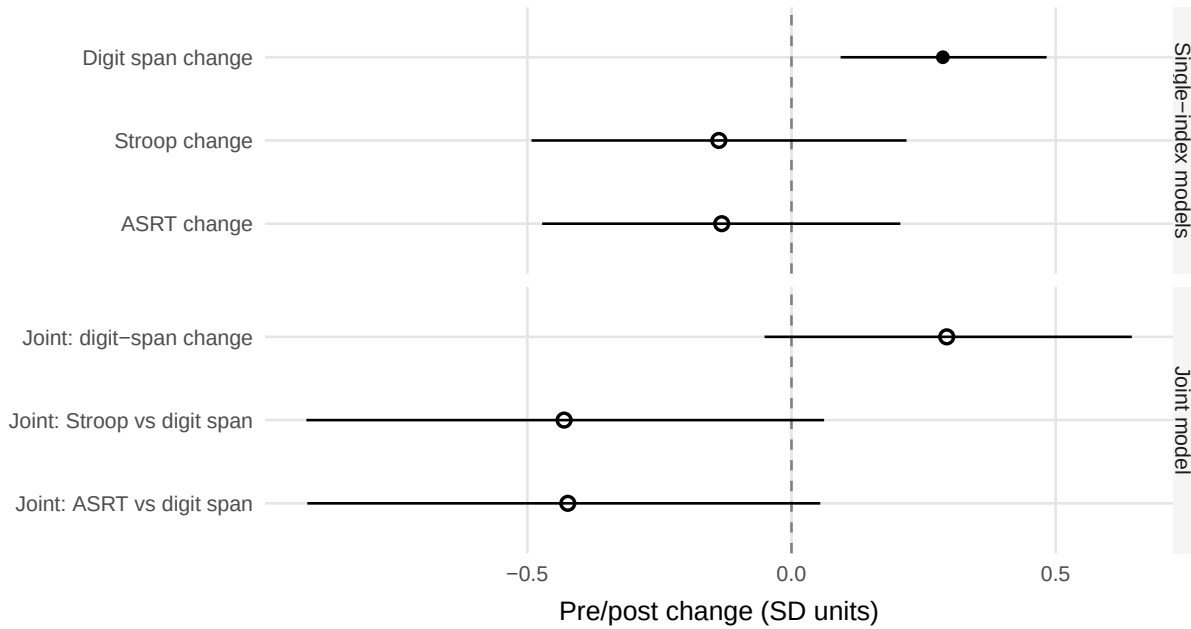
This study asked two questions: whether baseline cognitive and neural measures predict how well adults learn a new grammar (Part A), and whether executive function changes over the training period (Part B). The answer to the first is selective. Working memory predicted how well learners did, while inhibitory control, statistical learning and the resting-state measures added little, and the predictors acted on the level learners reached, leaving their rate of change unexplained. The answer to the second falls on the same capacity: working-memory span was the one index that rose over training.

Predictors of overall accuracy and learning rate

Working memory was the clearest cognitive predictor of how well learners did, credibly so for gender agreement and in the same direction, though less certainly, when the properties were pooled. This fits the literature placing working memory as the best-evidenced cognitive correlate of second-language outcomes (Linck et al., 2014). Inhibitory control and statistical learning,

Figure 5

Pre/Post (Session 1 to 5) Change in the Executive-Function Indices



Note. Points are single-index session effects and, for the joint model, the digit-span change and the contrasts of each measure's change against digit span (the reference level). Filled points mark effects whose 95% credible interval excludes zero; open points do not.

whose evidence base is weak or contested (Chen et al., 2022; Huensch, 2024; Linck & Weiss, 2015; Zhou et al., 2024), added little. Because digit span taps storage and not the executive or complex span that carries most of the working-memory–language association (Linck et al., 2014), a complex-span measure might well have shown a larger association. If so, the digit-span estimate is unlikely to overstate the working-memory contribution. Digit span is also the weaker of the phonological short-term memory tasks (Kurokawa, 2026, on vocabulary outcomes). That is a conjecture about what another measure would have shown, and it does not bound the estimate reported here.

The baseline resting-state measures were largely uninformative. Only the individual alpha frequency, a stable, trait-like measure (Grandy, Werkle-Bergner, Chicherio, Schmiedek, et al., 2013) related to general cognitive ability (Grandy, Werkle-Bergner, Chicherio, Lövdén, et al., 2013; Klimesch, 1999), hinted at a positive association with overall accuracy, and only in the

gender-agreement model, not the pooled one. It stands as a lead for confirmatory work, and one to be held lightly. In the same stratum, the projection-predictive search ranked the individual alpha frequency last, and the estimate is prior-sensitive (see *Priors*). In a small number of recordings, that measure is also a boundary value with no resolved peak behind it (see *Cognitive and neural indices*). The null for the band powers contrasts with reports that resting beta and low-gamma power predict L2 learning *rate* (Prat et al., 2016, 2019). We did estimate the rate analogue. The band-by-session interactions are in the model and none was credible, though under weakly informative priors their intervals are wide enough not to exclude effects of the published size. The divergence therefore tracks our posterior reduction where theirs was right-temporoparietal, together with our conservative priors and our precision. The two sets of results differ in scope without contradicting each other. On the gamma half of that comparison, we can say nothing at all. Our 30–45 Hz band lies above the 30 Hz offline low-pass, so the measure is not a valid index of gamma-band power and its null is uninformative by construction (see *Cognitive and neural indices*). Testing the low-gamma part of that result (Prat et al., 2016, 2019) would need a recording made without a 30 Hz low-pass and with the means to separate myogenic from neuronal activity. The aperiodic re-parameterisation sharpens the neural null. Had the flat band coefficients merely reflected the conflation of oscillatory power with the $1/f$ background (Donoghue et al., 2020), de-confounding them should have uncovered signal. That is not what happened. The aperiodic exponent and offset entered the selection well beyond the suggested size. Under participant-grouped cross-validation, neither characterisation predicted a held-out learner's trajectory detectably better than the other. Against a largely cross-sectional literature relating the aperiodic exponent to cognitive function, this bounds the question without answering it. At baseline, in this sample, the exponent carried no detectable incremental signal for a longitudinal training outcome, a test cross-sectional designs cannot run. That the two characterisations could not be distinguished at this sample size also means the band-power nulls above are not obviously attributable to periodic/aperiodic conflation, though a comparison this blunt (see the minimum detectable difference quoted in the Results) cannot establish that they are equivalent.

The predictors acted on the *level* learners reached, with no detectable purchase on the *rate*. The predictor-by-session interactions were weak, with the statistical-learning-by-session term the only one approaching credibility in the raw-band models, and only in the gender-agreement model. That model's rate is the least reliably estimated quantity here (reliability .42), so, per the rule stated in the Results, we do not interpret it. In the aperiodic re-parameterisation of that stratum, the adjusted-alpha-by-session term comes at least as close, and we do not interpret it either. The attainment-not-rate asymmetry is not an artefact of the priors' design, a small informative mass on attainment shrunk towards zero on rate, since it held when the rate terms were re-fit under weakly informative priors (see *Priors*).

The overall pattern is the one a circumscribed advantage would produce. Had a broad aptitude driven learning, it should have appeared in the statistical-learning index, which is reliably measured and was assessed and modelled identically. In the event, the signal was concentrated in working memory, and the projection-predictive selection retained nothing beyond it. The neural and inhibitory-control comparisons carry much less weight in that argument than the modelling alone suggests: the low-frequency bands are weakly identified, the gamma predictor is not a valid measure, and the Stroop index is too unreliable to have registered an association (see *Limitations and future directions*). The contrast the data can draw is therefore between working memory and statistical learning, the two indices measured well enough to be compared.

The joint model's chief advantage is that it separates attainment from rate. Estimating the by-participant intercept and slope inside one multilevel model, with their interactions with the predictors, propagates the uncertainty a two-stage slope analysis would discard ([Gelman, 2005](#); [Humberg et al., 2024](#)). The by-participant level was estimated with high reliability and the pooled learning rate less so (.75 against .84), with the gender-only rate more modest again (.42), so rate-based claims from that model warrant more caution ([Hedge et al., 2018](#)). This asymmetry bears on how aptitude findings indexed by a per-learner slope should be read. If the rate is in general the less reliably estimated quantity (as it was here even under trial-level joint modelling, the most favourable case), then slope-indexed predictor effects will be attenuated and unstable in

proportion to that unreliability. Disagreement among them is then to be expected ([Hedge et al., 2018](#); [Humberg et al., 2024](#)).

Change in executive function over training

A behavioural change also emerged over training. Working-memory span rose between the first and fifth sessions, while inhibitory control and statistical learning did not. This comparison lacks a no-training control group, so on its own a pre/post gain is open to the familiar threats of an uncontrolled retest: practice effects, regression to the mean, maturation and expectancy ([Barnett et al., 2005](#); [Calamia et al., 2012](#); [Shadish et al., 2002](#)). Two features of the analysis bound those threats. First, the gain appeared concentrated in one index. Of three indices measured and retested identically, only working memory moved. In a joint model across the battery, the pre/post change was larger for digit span than for the co-administered Stroop and statistical-learning measures, in the same direction and with probabilities of direction of 95.7% and 95.7%. Those specificity contrasts do not themselves clear the credibility threshold, since their 95% intervals include zero, and the joint-model digit-span change does not either, so the concentration remains provisional. The two co-administered measures nonetheless serve as internal controls. The threats of an uncontrolled retest act broadly across a battery, so a change concentrated in a single index is not what they predict. Both controls are imperfect. The Stroop index is of very low internal consistency here, which leaves it a noisy comparator, and its polarity runs opposite to digit span's, which inflates that contrast (see *Part B: change in executive function over training*). Second, that same working-memory measure was the one baseline predictor of overall accuracy, so the predictive and pre/post analyses converge on it. None of this licenses a causal claim. Practice effects are not uniform across tasks ([Scharfen et al., 2018](#)), and even active control conditions do not fully rule out differential expectancy ([Boot et al., 2013](#); [Simons et al., 2016](#)), so attributing the gain to the multilingual training specifically would require a matched no-training group. The result is consistent with a training-related, working-memory-specific change, and no more than that.

Limitations and future directions

The first limitation is precision. Although the models are fitted to tens of thousands of trials, the predictor effects are *between*-participant quantities, so their effective sample size is the number of participants. Only 56 participants contribute between-participant variance to the raw-band model and 50 to the aperiodic variant, and eighteen between-participant coefficients (nine main effects and nine interactions with time) are estimated from them. That is why even the clearest predictor effect is reported with a wide interval, and it bounds how strongly any single predictor claim can be made. Effects that do clear a 95% interval at this sample size are also, on average, overestimates of their true magnitude, so the working-memory coefficient should be read as evidence of a positive association, not as a calibrated estimate of its size.

A second limitation is that the study is silent on *neural* plasticity. Resting-state EEG was recorded only at baseline, so the rs-EEG measures bear only on the prediction question. Establishing whether intensive multilingual training reshapes resting-state activity would need a matched post-training recording, and the behavioural pre/post comparison reported above cannot stand in for one. A change in working-memory span is a change in what the task measures, and says nothing about whether the resting rhythms underlying that capacity have shifted.

A third limitation is the reliability of the predictors themselves, which bounds any association they can enter into. Because reporting the reliability of the outcome while leaving the predictors unquantified would be one-sided, we estimated the internal consistency of each baseline index by permutation split-half with a Spearman–Brown correction. The result separates two measures that are usually discussed together as difference scores ([Hedge et al., 2018](#)). Statistical learning is estimated highly reliably (.95) and working-memory span adequately so (.76), but Stroop interference is not. Its split-half reliability is .16, which is close to zero. Almost none of its between-participant variance is systematic, so the null association between inhibitory control and learning reported here is uninformative, and no association could have been detected with this measure however large the true effect. That applies to Part A and to Part B, where the Stroop index is a noisy internal control on the pre/post comparison. The statistical-learning null

carries no such excuse. The ASRT is estimated highly reliably, so its near-zero coefficient cannot be put down to measurement error. Even so, failing to detect an association is not the same as demonstrating its absence. Under a prior centred above zero, the pooled estimate is $b = 0.01$, 95% CrI $[-0.25, 0.27]$, $p_d = 53.4\%$. The interval's upper bound is close to the working-memory median we treat as our clearest effect, so these data cannot exclude a statistical-learning association of the size they attribute to working memory. What they establish is that no such association was detected on a well-measured index, which is why it, and not the neural or inhibitory-control nulls, is the comparator the specificity argument rests on.

A further scope condition is that no language-specific aptitude test was administered. All predictors index domain-general capacities, so the contrast our data can draw is between a single circumscribed capacity and a broad domain-general advantage. It does not extend to the language-specific aptitude of the Modern Language Aptitude Test (MLAT) and LLAMA tradition ([S. Li, 2015](#)). Whether such an aptitude measure would outpredict working memory here is an open question.

Finally, the artificial-language design trades ecological validity for control. The gain is a grammar no participant has met before, so prior exposure cannot confound the learning curve, and that control is what makes the individual-difference question tractable at all. The cost is that the three properties enter at different points and differ in difficulty, so the pooled trajectory averages over learners who are at different stages of different problems. The pooled analysis is therefore best read alongside the gender-only one, restricted to the single property sampled from the first grammar session onwards, which the predictor pattern survived.

Conclusion

Two lines of evidence point to working memory. It was the one baseline index that predicted how much grammatical knowledge learners acquired, and the one index that rose across the training period. Statistical learning, measured at least as reliably, did neither, and the inhibitory-control measure was too unreliable here to test either way. Both results are therefore about a single capacity, though the evidence that the pre/post gain is confined to it stops short of

being decisive.

Data and code availability

All extraction, modelling and reporting scripts, the computational environment specification and the machine-written summary tables that populate this manuscript are available in the project repository at <https://github.com/pablobernabeu/LESS-Project>, and are archived on OSF together with the online supplementary material. That supplement holds the baseline predictor distributions and the resting-state spectra, which are descriptive companions to the analyses reported here and are not needed to interpret them. The environment specification has two parts, because no single artefact covers the whole of it: an `renv` lockfile pinning the R packages of the fitting library, snapshotted on the cluster, and a provenance record written at fit time. The compiler and the optimisation pin described above belong to neither, being properties of the cluster module, which is why they are stated in prose. The provenance record covers the software versions in force when the Part A predictor models were fitted, so those fits can be traced to the environment that produced them. The pre/post models, the projection-predictive selection and the cross-validation comparisons run from the same environment, but they write no record of their own. That record has not yet been written for this paper, which is why the version fields under *Sampler*, *diagnostics* and *inference* print as pending. They populate from it automatically once the Part A predictor models are next fitted.

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Appendix A

Convergence diagnostics

Every model whose posterior estimates are reported in the main text was assessed against the same convergence criteria. Table A1 gives the realised per-model values behind the summary in the Results. The common-sample refits behind the cross-validation comparisons are not listed. They exist only to score the two parameterisations on identical observations, carry no reported coefficient and do not pass through the diagnostic artefact.

Table A1

Convergence Diagnostics for Every Model Whose Posterior Estimates Are Reported

Model	max R-hat	min ESS (bulk)	min ESS (tail)
Part A: pooled	1.002	1711	3338
Part A: pooled (weak priors)	1.001	7296	13915
Part A: gender agreement	1.003	2058	3412
Part A: gender agreement (weak priors)	1.001	7865	14510
Part A: pooled, aperiodic	1.001	7399	12996
Part A: gender agreement, aperiodic	1.001	6642	11246
Part B: digit span	1.001	5206	9883
Part B: Stroop interference	1.001	6068	10580
Part B: ASRT learning	1.001	4827	8724
Part B: joint across the three indices	1.001	11240	15912
Part B: joint across the three indices, sign-aligned	1.001	7486	10552

Note. Columns give the maximum rank-normalised split R-hat and the minimum bulk and tail effective sample sizes, each taken over all parameters. Part A models come first, then the Part B pre/post models. The weak-prior rows are the prior-sensitivity refits described under Priors.

Appendix B

Reliability of the baseline predictors

Table B1 gives the split-half internal consistency of the three baseline cognitive indices. The values, and what they imply for the inhibitory-control null, are discussed under *Limitations and future directions*.

Table B1

Internal-Consistency Reliability of the Three Baseline Cognitive Predictors at Session 1

Predictor	Participants	Mean half-test r	Reliability (Spearman-Brown)	Random splits
Working memory (digit span)	63	.61	.76	500
Inhibitory control (Stroop interference)	62	.09	.16	500
Statistical learning (ASRT)	63	.90	.95	500

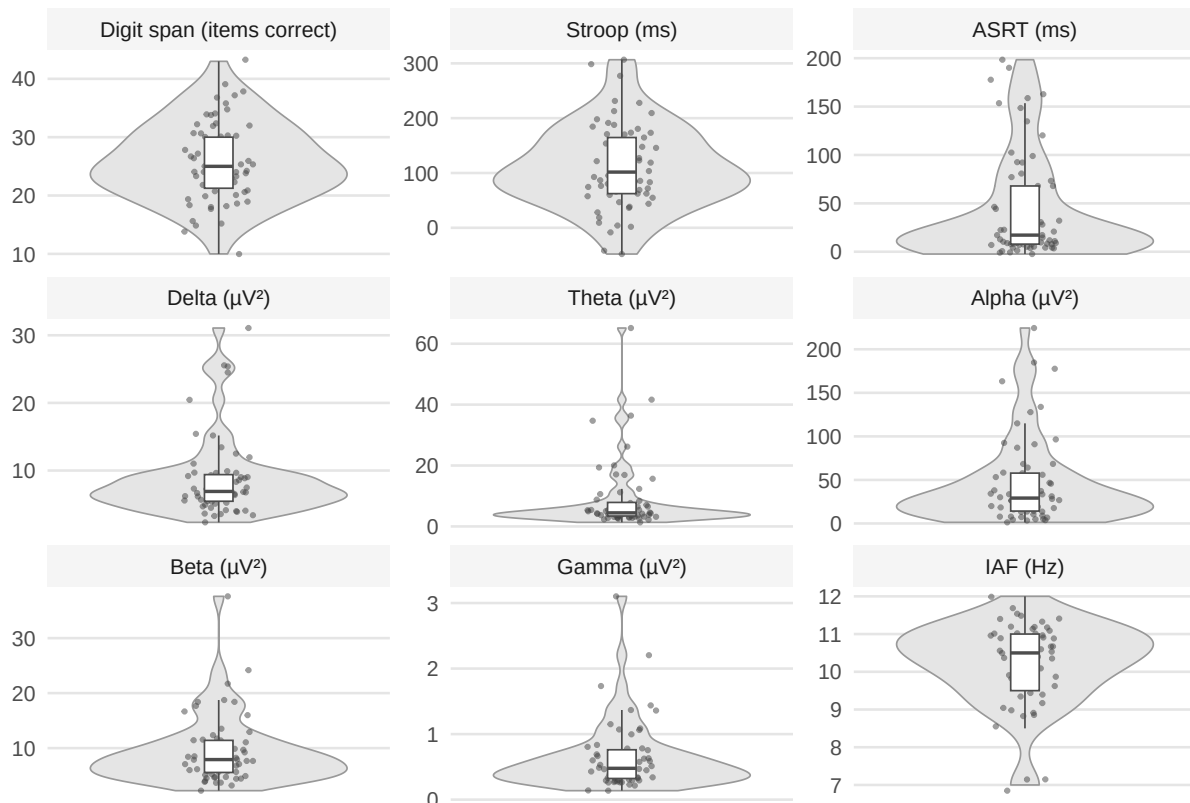
Note. Reliability is estimated by permutation split-half: each participant's trials are repeatedly split in half, the index is recomputed on each half, the halves are correlated across participants, and the mean correlation is corrected to full test length by the Spearman-Brown formula. Two of the three indices (Stroop interference, ASRT learning) are difference scores.

Appendix C

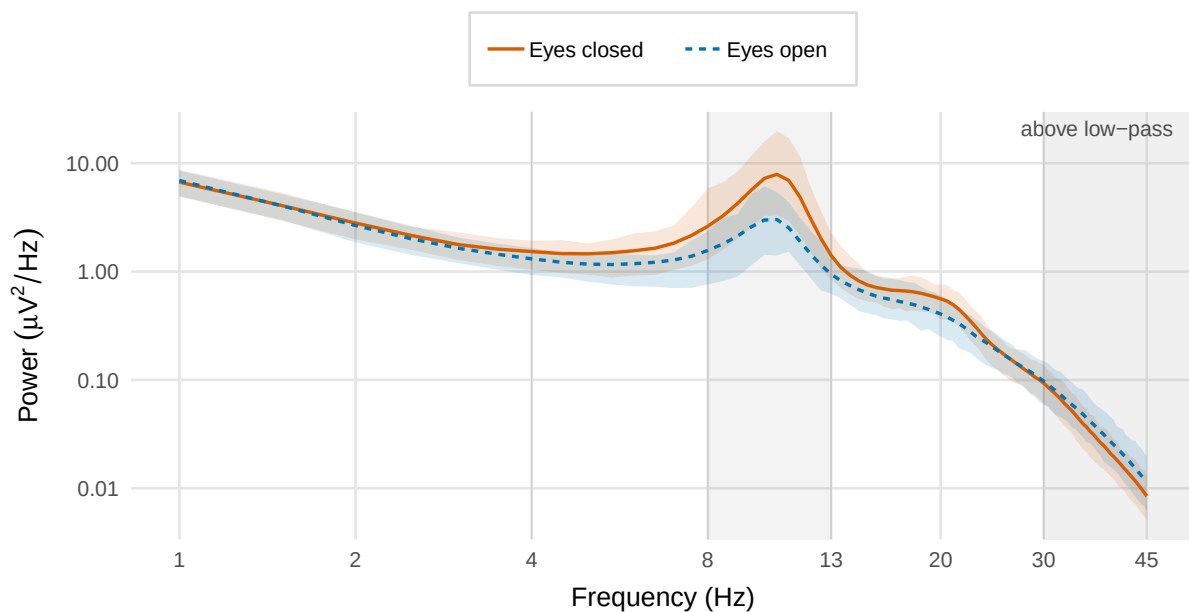
Supplementary materials

The two displays below are supplied to the journal as online supplementary material and are deposited with the analysis code in the project's OSF archive. They are descriptive companions to the intercorrelations and numerical summaries in the main text, so they are not required to interpret the findings, and they fall outside the five figures of the article proper. Figure [C1](#) shows the distribution of each baseline predictor over the stored records for that measure, and Figure [C2](#) the resting-state spectra behind the band-power and individual-alpha-frequency predictors.

Figure C1
Distributions of the Baseline Predictors



Note. Violins with individual points show working memory (digit span), Stroop interference and ASRT statistical-learning scores, and the eyes-closed resting-state band powers and individual alpha frequency. Each panel is drawn from the stored records for that measure, so the number of points differs between panels and exceeds the model samples reported under Participants. Difference-score predictors (Stroop, ASRT) are shown on their raw scale, and each panel's unit is given in its label. Box plots mark the median and interquartile range. The absolute band powers are strongly right-skewed. They nonetheless enter the models z-scored on this raw scale, with no transform applied. Points at the 7 Hz floor of the individual-alpha-frequency panel are values at the lower bound of the estimator's search range, not necessarily measured peaks (see Cognitive and neural indices).

Figure C2*Grand-Average Resting-State Power Spectral Density Over the Eight Occipito-Parietal Electrodes*

Note. Eyes-closed is shown against eyes-open. Both axes are logarithmic, so the aperiodic $1/f$ background appears as a straight line. The curve is the geometric mean across every recorded resting-state spectrum, a wider frame than the fitted models use, and the shaded ribbon is the interquartile range. Grey lines mark the interior delta/theta/alpha/beta/gamma band edges (4, 8, 13, 30 Hz), and the alpha band (8–13 Hz) is highlighted. The darker region above 30 Hz lies beyond the offline low-pass corner: the nominal gamma band (30–45 Hz) falls entirely within it, which is why that predictor is not interpreted. The eyes-closed posterior alpha increase (alpha-blocking on eye opening) is the expected Berger effect, which serves here as a check on the recording and the reduction.